Artificial Selection

Exam code: 0610 & 0970

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Updated on 15 May 2025

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Help me understand

Selective Breeding

- Selective breeding means to **select individuals with desirable characteristics and breed them together**
- The process doesn't stop there though because it's likely that not all of the offspring will show the characteristics you want so **offspring that do show the desired characteristics are selected and bred together**
- This process has to be **repeated for many successive generations** before you can definitely say you have a **'new breed'** which will **reliably** show those selected characteristics in all offspring

Selective breeding in domesticated animals

- An example of selective breeding is dog breeders who select which dogs can mate together to increase the likelihood of puppies displaying desirable characteristics eg. coat colour

This has given rise to the many recognisable dog breeds of today, all the same species (*Canis familiaris*) and all descended from one breed

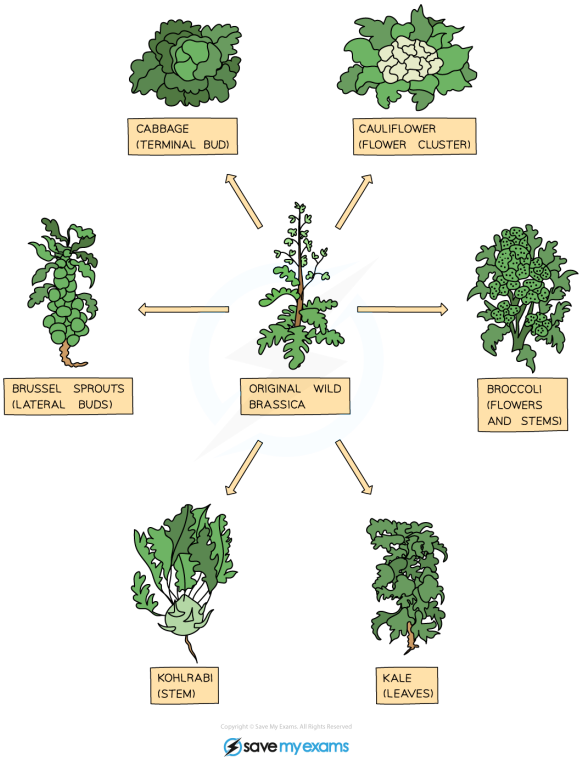
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Selective breeding in crop plants

- Plants are selectively bred by humans for the development of many characteristics, including:
  - **disease resistance** in food crops
  - **increased crop yield**
  - **hardiness to weather conditions** (e.g. drought tolerance)
  - better tasting fruits
  - large or unusual flowers
- An example of a plant that has been selectively bred in multiple ways is wild brassica, which has given rise to cauliflower, cabbage, broccoli, Brussels sprouts, kale and kohlrabi:



Selective breeding in crop plants

Artificial vs natural selection: Extended

RESULTS IN DEVELOPMENT OF POPULATIONS WITH FEATURES THAT ARE BETTER ADAPTED TO THEIR ENVIRONMENT AND SURVIVAL	RESULTS IN DEVELOPMENT OF POPULATIONS WITH FEATURES THAT ARE USEFUL TO HUMANS AND NOT NECESSARILY TO SURVIVAL OF THE INDIVIDUAL
USUALLY TAKES A LONG TIME TO OCCUR	TAKES LESS TIME AS ONLY INDIVIDUALS WITH THE DESIRED FEATURES ARE ALLOWED TO REPRODUCE

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